

In high cotton
A labourer at a cotton mill near Ahmedabad

AP

it was propelled by the private sector. Call it the 'double engine' growth of the milk sector. Indeed, the value of milk in India is more than the value of wheat, rice and sugarcane combined. An estimation of what is justly called the White Revolution can be had from the fact that in 1951, when India's milk production was 17mmt, the US had 53mmt. But by 2021, while US milk production hovered around 102mmt, India is touching 220mmt. And crucially, much of this is achieved by smallholders who have, on average, about four to five bovine animals (cows and buffaloes). It is the result of the institutional innovation that Verghese Kurien, the 'milkman of India', introduced by bringing small milk producers together through milk cooperatives to create scale, and then linking that with modern processing (pasteurising and homogenising) and organised retailing in the megacities of India.

Interestingly, the Indian agriculture landscape has changed even more over the past two decades. The Vajpayee government in 2002-03 also took a bold decision to allow the first genetically modified crop (cotton) to be grown in India. With the introduction of **Bt cotton** in 2002-03, primarily led by the big private sector seed companies, India's cotton production catapulted from 13.6 million bales (one bale is 170 kg) in 2002-03 to 39.8 million bales in 2013-14, almost a 293 per cent increase in just 11 years. This is unmatched in the history of agriculture in India. This 'gene revolution' in cotton made India the largest producer and second-largest exporter of cotton in the world. This is the fourth big milestone.

FARMERS SHOULD GET UNHINDERED ACCESS TO THE BEST TECHNOLOGIES FROM ANYWHERE IN THE WORLD AND TO THE GLOBAL MARKETS

From 2010-11 onwards, Indian agriculture has been under rapid transformation. It is worth noting that from 2010-11 to 2019-20, the value of Indian poultry meat (at constant prices) has been growing at an unprecedented rate of 10.4 per cent per annum, that of the fishery at 7.8 per cent, pulses at 5.7 per cent, dairy at 5.5 per cent, horticulture at 4.9 per cent, eggs at 4.9 per cent, cotton at 4.6 per cent, sugarcane at 3.9 per cent, oilseeds at 3.3 per cent, and cereals at 3.1 per cent. All these annual growth rates need to be seen in relation to the population growth rate, which has been declining gradually over the years and stands at around 1 per cent today. This clearly indicates that the per capita availability of food and fibre has been increasing quite fast.

That India has gone from food deficiency to self-sufficiency—except for the notable exception of edible oils, where we still import 55 to 60 per cent of our needs—is itself surely one of the biggest achievements. Now, Indian agriculture is a **net exporting sector**. The fifth biggest achievement is that this success story was written by the smallholders of India whose average holding size is a meagre 1.08 hectare (as per the 2015-16 Land Census), duly supported by science, be it of Indian origin or imported. Remember, the seeds of the Green Revolution came from outside the country and so did of Bt cotton. Lesson for future change: farmers should get access to the best technologies from anywhere in the world. And if a policy environment is created wherein they can also access global markets unhindered, one can confidently say that the best of agriculture in India is yet to come. ■